Encoding and Feature Scaling (Notes)

1. Encoding

Definition

Encoding is the process of converting **categorical (text) data into numerical form** so that machine learning algorithms can understand it.

Most ML algorithms work only with **numerical data**, so categorical variables must be encoded.

Types of Encoding

1. Label Encoding

Each category is assigned a **unique integer value**.

Example:

Color	Encoded
Red	0
Blue	1
Green	2

Python Code:

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
data['Color'] = le.fit_transform(data['Color'])
```

Advantages

- Simple and fast
- Requires less memory

Disadvantages

- Model may assume order between categories

Use Case

- Ordinal data (Low, Medium, High)

2. One-Hot Encoding

Creates **separate binary columns** for each category.

Example:

Color	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1

Advantages

- Removes ordinal relationship

Disadvantages

- Increases number of columns

Use Case

- Nominal data (City, Gender, Product)

2. Feature Scaling

Definition

Feature Scaling is a technique used to **normalize the range of independent variables or features** in a dataset.

Different features may have **different units and ranges**, which can affect model performance.

Types of Feature Scaling

1. Standardization (Z-score Normalization)

Transforms data so that:

- Mean = 0
- Standard Deviation = 1

Formula:

Z=(X−μ)/σZ = (X - \mu) / \sigmaZ=(X−μ)/σ

Used in:

- Logistic Regression
- SVM
- Neural Networks
- KNN